x^2 The quadratic formula

QUADRATIC EQUATIONS CAN BE SOLVED USING A FORMULA.

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The quadratic formula

The quadratic formula can be used to solve any quadratic equation. Quadratic equations take the form $ax^2 + bx + c = 0$, where a, b, and c are numbers and x is the unknown.

▷ **A quadratic equation** Quadratic equations include a number multiplied by x^2, a number multplied by x, and a number by itself.

number that multiplies x^2

number that multiplies x

number with no x terms

$$ax^2 + bx + c = 0$$

▷ **The quadratic formula** The quadratic formula allows any quadratic equation to be solved. Substitute the different values in the equation into the quadratic formula to solve the equation.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

this means add or subtract

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Quadratic variations

Quadratic equations are not always the same. They can include negative terms or terms with no numbers in front of them ("x" is the same as "1x"), and do not always equal 0.

the values in the equation can be negative as well as positive

quadratic equations are not always equal to 0

$$-4x^2 + x - 3 = 8$$

when an x appears without a number in front of it, x=1

Using the quadratic formula

To use the quadratic formula, substitute the values for a, b, and c in a given equation into the formula, then work through the formula to find the answers. Take great care with the signs (+, –) of a, b, and c.

Given a quadratic equation, work out the values of a, b, and c. Once these values are known, substitute them into the quadratic formula, making sure that their positive and negative signs do not change. In this example, a is 1, b is 3, and c is –2.

the value of a is 1

the value of b is 3

the value of c is –2

$$x^2 + 3x - 2 = 0$$

$$x = \frac{-3 \pm \sqrt{3^2 - 4 \times 1 \times (-2)}}{2 \times 1}$$

substitute values from equation into the formula, keeping their signs the same